Lab Sheet 08

Load and Prepare the Data

We will use the built-in "mtcars" dataset for this lab. The dataset contains various measurements of car performance and design.

```
# Load the mtcars dataset
data("mtcars")
head(mtcars) # View the first few rows

# Standardize the data
mtcars_scaled = scale(mtcars)
head(mtcars_scaled) # View the scaled data
```

Perform PCA

We will use the prcomp() function to perform PCA on the scaled dataset.

```
# Perform PCA
pca_result = prcomp(mtcars_scaled, center = TRUE, scale. = TRUE)

# Summary of PCA results
summary(pca_result)

# View the rotation matrix (principal components)
pca_result$rotation
```

```r
# View the principal component scores
head(pca_result$x)
```

Visualize PCA Results

We will use the factoextra and ggplot2 packages to visualize the PCA results.

```r
# Install and load the required libraries
library(factoextra)
library(ggplot2)

# Scree plot to visualize variance explained by each principal component
fviz_eig(pca_result)

# PCA biplot
fviz_pca_biplot(pca_result, label = "var", addEllipses = TRUE)

# PCA individual plot
fviz_pca_ind(pca_result, addEllipses = TRUE)

# PCA variable plot
fviz_pca_var(pca_result)
```

Reconstruct Data Using Principal Components

If needed, we can reconstruct the original data using a subset of principal components.

```
# Reconstruct the data using the first two principal components

reconstructed_data = pca_result$x[, 1:2] %*% t(pca_result$rotation[, 1:2])

# Add the mean back to the reconstructed data

reconstructed_data = scale(reconstructed_data, center = -colMeans(mtcars),
scale = FALSE)

head(reconstructed_data)
```